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$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{x - \frac{\pi}{2}}{\cos x}$$

$$y = x - \frac{\pi}{2}$$

$$\begin{aligned} &= \lim_{y \rightarrow 0} \frac{y}{\cos\left(y + \frac{\pi}{2}\right)} \\ &= \lim_{y \rightarrow 0} \frac{y}{-\sin y} = - \lim_{y \rightarrow 0} \frac{y}{\sin y} \\ &= -1 \end{aligned}$$

References